

Vehicle System Dynamics

Youqun Zhao · Fen Lin

Vehicle System Dynamics

Theoretical Modeling and Application

Youqun Zhao
College of Energy and Power Engineering
Nanjing University of Aeronautics
and Astronautics
Nanjing, Jiangsu, China

Fen Lin
College of Energy and Power Engineering
Nanjing University of Aeronautics
and Astronautics
Nanjing, Jiangsu, China

ISBN 978-981-97-2018-7 ISBN 978-981-97-2019-4 (eBook)
<https://doi.org/10.1007/978-981-97-2019-4>

Jointly published with Science Press

The print edition is not for sale in China (Mainland). Customers from China (Mainland) please order the print book from: Science Press.

© Science Press 2024

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publishers, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publishers nor the authors or the editors give a warranty, express or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publishers remain neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Singapore Pte Ltd.

The registered company address is: 152 Beach Road, #21-01/04 Gateway East, Singapore 189721, Singapore

If disposing of this product, please recycle the paper.

Preface

Vehicle system dynamics is an emerging discipline that has developed in modern times, encompassing all the sciences related to automotive motion, with rich and extensive content. Specifically, vehicle system dynamics studies the mechanisms of longitudinal, lateral, and vertical motion of vehicles, as well as the dynamic characteristics and control technologies of various subsystems and assemblies of vehicles derived from these three movements.

With the rapid development of automotive electronic control technology, people are more and more aware of the importance of vehicle system dynamics. The current advanced active safety systems in automobiles, such as anti-lock braking system (ABS), electronic stability program (ESP), adaptive cruise control (ACC), acceleration slip regulation (ASR), electronic brake force distribution (EBD), etc., are based on vehicle dynamics as a theoretical basis for research and development and design. Therefore, for the undergraduate and graduate students majoring in vehicle engineering and the related engineering technicians who will be engaged in the research and development of vehicle engineering technology in the future, it is necessary to study the vehicle system dynamics systematically and deeply.

With the development of modern vehicle performance and speed, and the application of chassis control technology, the research content of vehicle system dynamics is more and more abundant. In the process of writing this book, a large amount of research and material collection are conducted to track the latest industry development status. Combined with the author's years of teaching and research experience in the industry, the latest vehicle handling inverse dynamics, non-pneumatic tire dynamics, vehicle state and parameter estimation methods, and driver-vehicle closed-loop control system dynamics methods are taken as the key chapters of this book. It fully reflects the current advanced technology status and development trend of vehicle system dynamics. In addition, this book systematically combs and classifies the knowledge points of vehicle system dynamics, forming a highly systematic framework structure, aiming to help readers deeply understand the knowledge points of vehicle system dynamics.

The research work of this book has been supported by the National Natural Science Foundation of China (52272397, 11672127, 11072106, 10902049), the

Doctoral Special Research Fund of Chinese Universities (20040287004), the Open Fund of China National Engineering Laboratory for High-Mobility Anti-riot Vehicle Technology (B20210017), the Fundamental Research Funds for the Central Universities (NP20220408, NP2020407, NS2020016, NP2018403, NT2018002, NP2016412, NS2015015), China Postdoctoral Foundation Special Funding Project (2017T100365), China Postdoctoral Foundation superficial Funding Project (2016M601799, 2012M521073), the Key Research & Development and Achievement Transformation Program of Wuhu (2023yf010), and China Jiangsu Postdoctoral Fund Project (1302020C). Here we express our sincere gratitude.

Chapter 7 and Sects. 3.3, 4.3, 6.3 and 6.4 were written by Prof. Lin Fen, and the rest of the book was written by Prof. Zhao Youqun.

In order to present a more complete and rich theoretical system to readers, some domestic and foreign literatures have been quoted to enrich the content of this book during the writing process, and we would like to thank the authors of these literatures.

Due to the limited level of the authors, there are unavoidable omissions in the book, please criticize and correct the majority of readers.

Nanjing, China

Youqun Zhao
Fen Lin

Introduction

This book covers the basic theoretical systems of vehicle system dynamics, based on the modeling and theoretical methods of longitudinal, lateral, and vertical system dynamics. The research results of pneumatic/non-pneumatic tire dynamics, vehicle handling inverse dynamics, vehicle state and parameter estimation method, new evaluation method of driver-vehicle closed-loop system handling dynamics, four-wheel steering vehicle motion stability and vehicle suspension control, and so on are introduced in detail. In the process of reading this book, readers will not only have a comprehensive understanding of the framework system of vehicle system dynamics, but also have a deeper grasp of the connotation and denotation of the discipline.

This book can be used as a course material for undergraduate and graduate students in the field of vehicle engineering. In addition, this book for the vehicle professional teaching and research personnel also has a good reference value.

Contents

1	Introduction	1
1.1	System and System Dynamics	1
1.2	Structure and Composition of Vehicle System	3
1.3	Research Contents and Methods of Vehicle System Dynamics	5
1.3.1	Research Contents	5
1.3.2	Research Methods	7
1.4	Theory and Research Methods of Vehicle Handling Inverse Dynamics	12
1.4.1	Principle of Handling Inverse Dynamic	12
1.4.2	Modeling and Solution of Handling Inverse Dynamics	14
	References	19
2	Mechanical Characteristics of Pneumatic and Non-pneumatic Tires	21
2.1	Construction of Tires	21
2.1.1	Construction of the Pneumatic Tire	21
2.1.2	Construction of the Non-pneumatic Mechanical Elastic Safe Wheel	22
2.2	Mechanical Characteristics of the Pneumatic Tire	22
2.2.1	Coordinate System of the Tire	22
2.2.2	Tire Cornering Phenomenon and Lateral Force-Lateral Angle Curve	23
2.2.3	Aligning Torque	26
2.2.4	Typical Mechanics Model of the Tire	28
2.3	Mechanical Characteristics of Non-pneumatic Wheels	31
2.3.1	NMESW Loading Mode and Working Principle	31
2.3.2	Rolling Mechanism of NMESW	32
2.3.3	Lateral Characteristic of NMESW	38
2.3.4	Camber and Cornering Characteristics of NMESW	40
	References	45

3	Vehicle Stability Control and Estimation of Road Friction	
	Coefficient Based on Vehicle Longitudinal Dynamics	47
3.1	Review on Vehicle Longitudinal System Dynamics	47
3.1.1	Significance of Research on Dynamic Characteristics of Vehicle Longitudinal System	47
3.1.2	Basic Problems in the Study of Dynamic Characteristics of Vehicle Longitudinal System	48
3.2	Vehicle Stability Control System	49
3.2.1	Summarize	49
3.2.2	Basic Principle	50
3.2.3	Control Method	50
3.2.4	Dynamic Model and Simulation	53
3.3	Estimation of Road Friction Coefficient Based on Longitudinal Dynamics	57
3.3.1	Slip-Slope Based Method for Estimating Friction Coefficient	57
3.3.2	Longitudinal Force Estimation	59
3.3.3	Slip Estimation	63
3.3.4	Slip-Slope Estimation Method Based on RLS	64
3.3.5	Virtual Test Verification	67
	Appendix	76
	References	80
4	Independent All-Wheel Drive Distribution Control and Estimation of Tire Effective Cornering Stiffness Based on Vehicle Lateral Dynamics	83
4.1	Review of Vehicle Lateral System Dynamics	84
4.1.1	Introduction of Evaluation Method Development for Vehicle Lateral System Dynamic Characteristics	84
4.1.2	The Significance of Research on Closed-Loop Lateral System Dynamics	87
4.1.3	Basic Problems in the Vehicle Lateral System Dynamics	88
4.2	Independent All-Wheel Drive Distribution Control	89
4.2.1	Conventional Four-Wheel Drive System	89
4.2.2	Differential-Based Torque Distribution Between Left and Right Wheels	90
4.2.3	Active Control of All-Wheel Torque	90
4.3	Estimation of Tire Effective Cornering Stiffness	92
4.3.1	Estimation of Tire Effective Cornering Stiffness Based on the Time Rate of Change of Acceleration	92
4.3.2	Feasibility Analysis of Estimation Method	96
4.3.3	Simulation Test Verification	99
	References	105

5	Evaluation of Active Safety Based on Driver-Vehicle Closed-Loop Control System Dynamics	107
5.1	Theoretical Basis	107
5.1.1	Driver-Vehicle–Road Closed-Loop System Model	107
5.1.2	Evolutionary Random Processes	111
5.1.3	Kronecker Algebra Basis	113
5.1.4	Second Moment Technique	115
5.2	Response Analysis Method of Driver-Vehicle–Road Closed Loop Control System	116
5.2.1	Existing Precise Integration Algorithm	117
5.2.2	Extension of Precise Integration Algorithms	119
5.2.3	Numerical Simulation Example	120
5.3	Evaluation Index of Vehicle Active Safety	122
5.3.1	Supplementary to the Closed-Loop Individual Evaluation Indexes	124
5.3.2	Supplement to the Open-Loop Individual Evaluation Indexes	125
5.3.3	Selection of Comprehensive Evaluation Index and Weighting Coefficient	126
5.3.4	Correlation Between Closed-Loop Comprehensive Evaluation Index and Open-Loop Comprehensive Evaluation Index	127
5.3.5	Numerical Simulation Example	128
5.4	Summary	131
	References	133
6	A New Evaluation Method for Driver-Vehicle Closed-Loop Handling System Dynamics	135
6.1	Maneuverability Evaluation of the Driver-Vehicle Closed-Loop System with Random Road Inputs	135
6.1.1	Driver-Vehicle–Road Closed-Loop System Model with Random Road Input	136
6.1.2	Self-Spectral Density of Effective Road Input	138
6.1.3	Time Domain Analysis of Driver-Vehicle–Road Stochastic Closed-Loop System Response	139
6.1.4	Numerical Simulation Example	141
6.1.5	Virtual Input Algorithm	144
6.1.6	Conclusion	148
6.2	Influence of Driver’s Dynamic Characteristics on Vehicle Handling Safety	148
6.2.1	Driver-Vehicle–Road Stochastic Closed-Loop System Model	149
6.2.2	General Stochastic Perturbation Method for Response Analysis of Stochastic Closed-Loop Systems	153

6.2.3	The Influence of Driver's Dynamic Characteristics on Vehicle Active Safety	154
6.2.4	Numerical Simulation Example	155
6.2.5	Conclusion	164
6.3	Motion Stability Analysis of Four-Wheel Steering Vehicle	164
6.3.1	Motion Stability Theory	165
6.3.2	Establishment of Driver-Four-Wheel-Steering-Vehicle Closed-Loop System Model	166
6.3.3	Motion Stability Analysis of Four-Wheel Steering Vehicle	167
6.3.4	Conclusion	169
6.4	Application of Matrix Perturbation Method in the Stability Analysis of Four Wheel Steering Vehicle	170
6.4.1	Matrix Perturbation Method for Eigenvalue Problems of Asymmetric Matrices	170
6.4.2	Influence of Vehicle Parameters on Motion Stability	173
6.4.3	Conclusion	174
	References	176
7	States and Parameters Estimation in the Vehicle System Dynamics	179
7.1	Vehicle Side-Slip Angle Estimation	179
7.1.1	Driver-Vehicle Closed Loop System Model	180
7.1.2	Estimation of the Side Slip Angle Based on the Radial Basis Function Neural Network	185
7.1.3	Estimation of the Side Slip Angle Based on Adaptive Kalman Filter	189
7.1.4	Comparison of Two Estimation Methods of Side Slip Angle	191
7.2	Vehicle States Estimation Based on UKF Algorithm	196
7.2.1	Nonlinear Vehicle System with Noise	196
7.2.2	UKF Algorithm for Vehicle States Estimation	197
7.2.3	Test Verification of Vehicle UKF Estimation Methods	201
7.2.4	Comparison of UKF and EKF in Vehicle States Estimation	210
7.3	Vehicle States Estimation Based on DEAKF	212
7.3.1	Nonlinear Seven Degrees of Freedom Vehicle Dynamics Model	212
7.3.2	Dual Extended Adaptive Kalman Filter Algorithm	215
7.3.3	Three Methods of Vehicle States Estimation and Their Virtual Test Verification	219
7.3.4	Real Vehicle Test Verification	227

7.4	Vehicle States Estimation Based on Fuzzy Logic	228
7.4.1	Fuzzy Kalman Filter and S-AKF Algorithm for State Estimation	228
7.4.2	Virtual Test Verification	232
7.5	Parallel Estimation of Vehicle States and Parameters Based on EKF and RLS	239
7.5.1	Nonlinear Vehicle Dynamic Model	239
7.5.2	EKF and RLS Parallel Algorithm	241
7.5.3	Virtual Test Verification and Analysis	244
7.5.4	Real Vehicle Test Verification and Analysis	247
	References	251
8	Vehicle Handling Inverse Dynamics	253
8.1	Optimal Control Theory and Solution Methods	253
8.1.1	Optimal Control Theory	253
8.1.2	Optimal Control Solution Method	254
8.1.3	Sequential Quadratic Programming Algorithm	263
8.2	Research on Linear Vehicle Angle Input Recognition	265
8.2.1	Vehicle Angular Input Steering Motion Model	265
8.2.2	Simulation Results	268
8.3	Research on Linear Vehicle Force Input Recognition	271
8.3.1	Vehicle Force Input Steering Motion Model	271
8.3.2	Simulation Results	272
8.4	ADAMS/Car Model Building and Test Verification	273
8.4.1	ADAMS/Car Vehicle Modelling	274
8.4.2	Closed-Loop Control Analysis	275
8.4.3	ADAMS/Car Verification	276
8.5	Summary	276
	References	277
9	Road Roughness, Evolution Random Response and Ride Comfort Based on Vehicle Vertical Dynamic	279
9.1	Development Status and Trends	280
9.2	Study on Frequency Domain Inverse Dynamics of Road Roughness	281
9.2.1	Identification of Four-Degree-of-Freedom Vehicle Road Roughness in Frequency Domain	281
9.2.2	Frequency Domain Identification of Seven-Degree-of-Freedom Vehicle Road Surface Roughness	286
9.2.3	Virtual Test Verification	290
9.3	Study on Time Domain Inverse Dynamics of Road Roughness	297
9.3.1	Identification of Four-Degree-of-Freedom Vehicle Road Roughness in Time Domain	297

9.3.2	Identification of Seven-Degree-of-Freedom Vehicle Road Roughness in Time Domain	299
9.3.3	Virtual Test Verification	302
9.4	Prediction of Evolution Random Response of Vehicle Excited by Road Surface	309
9.4.1	Vehicle Vertical Dynamics Model	309
9.4.2	Response of Vehicle Vertical Vibration	311
9.4.3	Numerical Simulation Case	313
9.4.4	Conclusions	315
9.5	Optimal Design of Vehicle Ride Comfort Based on Bionic Lizard Co-evolution Algorithm	316
9.5.1	Principles of Bionic Lizard Co-evolution Algorithms	316
9.5.2	Key Technologies of Bionic Lizard Co-evolution Algorithms	318
9.5.3	Simulation Results and Analysis	323
9.6	Hydro-Pneumatic Suspension Control Matched with Mechanical Elastic Wheels	331
9.6.1	Vehicle Ride Simulation Model	332
9.6.2	Optimization of Design Parameters of Hydro-Pneumatic Suspension	334
9.6.3	Analysis of Optimization Results	336
	References	338